# Controlled numerical and missingness shift

Review qualified findings and the completed analytical scope. No qualifying issue means none within completed definitions and bounds.

Execution: completed · Backend: duckdb_local. A clean result means no qualifying issue within completed definitions and bounds.

## 1. Summary

2 qualified review findings across 2 source(s). Comparisons have no composite score or risk band.

## 2. Actions

**Moderate** · current

### Substantial column missingness

Confirm the selected missing-value policy and investigate collection coverage before imputing or excluding values.

observation. Evidence does not establish a defect, causality, or leakage.

**Moderate** · reference:current

### Supported input or annotation populations differ

Review source provenance and the declared use.

observation. Evidence does not establish a defect, causality, or leakage.

## 3. Sources and declarations

| source | rows | columns | sha256 | task |
|---|---|---|---|---|
| reference | 240 | 5 | 628c26dfe624e4b3b99d9250bae8f3592329fbd8de789d80b29ade3622c32286 | classification |
| current | 240 | 5 | 3da23ac1550971f2cdb20e0c372e496d3c53eadc5cb62c7cc3317c559973b3ff | classification |

Alignment is case-sensitive and declaration-bound. Identifier/key strings preserve decoded lexical identity, including leading zeros and whitespace. Effective parsing, annotation, semantic, mapping and key declarations are retained in JSON.

## 4. Overlap and schema evidence

| pair | added | removed | storage or role changes |
|---|---|---|---|
| reference:current | None | None | 1 |

Overlap counts are participating rows and grouped multiset intersections. Shared observations do not prove leakage; absence applies only to completed definitions.

## 5. Feature populations and missingness

| source | column | role | storage | missing | missing percent | nonfinite | distinct |
|---|---|---|---|---|---|---|---|
| reference | id | metadata | VARCHAR | 0 | 0.0 | 0 | 240 |
| reference | x | feature | BIGINT | 0 | 0.0 | 0 | 240 |
| reference | z | feature | BIGINT | 0 | 0.0 | 0 | 240 |
| reference | channel | feature | VARCHAR | 0 | 0.0 | 0 | 2 |
| current | id | metadata | VARCHAR | 0 | 0.0 | 0 | 240 |
| current | x | feature | BIGINT | 0 | 0.0 | 0 | 240 |
| current | z | feature | DOUBLE | 60 | 25.0 | 0 | 180 |
| current | channel | feature | VARCHAR | 0 | 0.0 | 0 | 2 |

| pair | column | left rows | right rows | missing change pp | nonfinite change pp | TV | CDF | Wasserstein 1 | block |
|---|---|---|---|---|---|---|---|---|---|
| reference:current | x | 240 | 240 | 0.0 | Unavailable | Unavailable | 0.416667 | 100.0 | None |
| reference:current | z | 240 | 240 | 25.0 | Unavailable | Unavailable | 0.25 | 60.0 | None |
| reference:current | channel | 240 | 240 | 0.0 | Unavailable | 0.0 | Unavailable | Unavailable | None |
| reference:current | label | 240 | 240 | 0.0 | Unavailable | 0.0 | Unavailable | Unavailable | None |

Missingness changes are signed percentage points. Distributions use observed categorical or finite quantitative values. Wasserstein-1 retains original units. Counts, full-population effects and separately qualified optional tests appear in JSON.

## 6. Targets, constraints and segments

| source | column | role | storage | missing | missing percent | nonfinite | distinct |
|---|---|---|---|---|---|---|---|
| reference | label | target | VARCHAR | 0 | 0.0 | 0 | 2 |
| current | label | target | VARCHAR | 0 | 0.0 | 0 | 2 |

Annotation expectation is independent of observed validity. Complete and partial vectors, declared requirements, subgroup mix and conditional populations retain their own denominators. Correlation, IQR candidates and conditional differences do not establish causality, fairness or anomaly correctness.

## 7. Methodology and coverage

Accepted task-family support does not mean exhaustive analytical searches. Planned limits complete bounded scope; omitted detail and unsearched scope remain explicit.

Method m4_local_2026_10_10. Parsing scalable_csv_v1. DuckDB 1.5.6. Searches are deterministic and bounded; required structural and configured overlap populations use every eligible record.

Resource defaults and runtime safeguards are engineering controls, not capacity guarantees. Disk planning estimates are conservative estimates, not upper bounds. Runtime 0.36 seconds; sampled peak worker RSS 119799808 bytes; recorded peak workspace 65536 bytes.

Parsed equality does not establish byte-identical records or proven leakage.

Effects and support describe supplied populations; conditional evidence is descriptive.

Search bounds and detail omissions remain visible in coverage.

## 8. Appendix and retained scope

| check | status | population | unsearched | excluded columns |
|---|---|---|---|---|
| reference.structure | completed | all eligible records | 0 | |
| reference.annotations | completed | all eligible records | 0 | |
| reference.repeated_inputs | completed | all eligible records | 0 | |
| reference.annotation_cohorts | completed | all eligible records | 0 | |
| current.structure | completed | all eligible records | 0 | |
| current.annotations | completed | all eligible records | 0 | |
| current.repeated_inputs | completed | all eligible records | 0 | |
| current.annotation_cohorts | completed | all eligible records | 0 | |
| reference:current.schema | completed | all eligible records | 0 | |

| check | status | population | unsearched | excluded columns |
|---|---|---|---|---|
| reference:current.populations | completed | observed categorical or finite quantitative values; missing/non-finite counts separately | 0 | |

Excluded-column previews show at most three shortened names; complete sets are retained in JSON. Human detail omissions, unsearched analytical scope, blocked methods and failed execution are distinct. JSON preserves all retained computed records. Generated 2026-10-10T17:41:09.211111+00:00.